

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: G. S. Sanyal School of Telecommunications

2. Subject Name: Principles of Communication Engineering
Credits: 3

L-T-P: 3-0-0

3. Pre-requisites:

4. Syllabus and reference books:

Syllabus:

Introduction, Narrowband Signals and Systems, Baseband, passband, Carrier Modulation, IF and RF sections of Communication transceivers, Wireline digital transmission schemes, Elements of Wireless Communications, Case studies.

Reference Books:

1. Communication Systems by Simon Haykin
2. Principles of Communication Systems by Herbert Taub and Donald L. Schilling,
3. Communication Systems by A. Bruce Carlson,
4. Modern Digital and Analog Communication Systems by B. P. Lathi and Zhi Ding
5. Communication Systems Analysis and Design by Harold P. E. Stern and Samy A. Mahmoud
6. Modern Communication Systems – Principles and Applications by Leon W. Couch II
7. Modern Communication Circuits by Jack Smith

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction (Popular examples; Wire line and Wireless media; Frequency spectrum; Bandwidth; Radio Wave Propagation; Path loss and Friis' Transmission Model; Trade-offs in Communication system design)	03
2.	Representation of Narrowband Signals and Narrowband Systems	03
3.	Random signals – analog and digital; Noise	03
4.	Baseband transmission vis-a-vis Carrier Modulation	02
5.	Analog Carrier Modulation and Demodulation Schemes – Amplitude Modulation (Basic Concepts of AM, DSB-SC, SSB-SC, VSB, circuit schematic, performance assessment)	06

6.	Angle Modulation and Demodulation Schemes (Basic Concepts of narrow band FM, Wideband FM, Angle Modulation, circuit schematics, performance assessment)	05
7.	Analog Pulse Modulation schemes (Basic Concepts of PAM, PWM, PPM and circuit schematics)	03
8.	IF and RF Sections of Communication Transceivers – Circuit Schematics	03
9.	Wireline Digital Transmission Schemes - copper cable and optical fiber as transmission media	03
10.	Elements of Wireless Digital Communications	06
11.	Case Studies	03
Total number of hours		40